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```
# Python 3: List comprehensions
>>> fruits = ('Banana', 'Apple', 'Lime')
>>> loud_fruits = [fruit.upper() for fruit in fruits]
>>> print(loud_fruits)
['BANANA', 'APPLE', 'LIME']

# List and the enumerate function
>>> list(enumerate(fruits))
[(0, 'Banana'), (1, 'Apple'), (2, 'Lime')]
```

Compound Data Types

Lists (known as arrays in other languages) are one of the compound data types that Python understands. Lists can be indexed, sliced and manipulated with other built-in functions. [More about lists in Python 3](#)

- 1
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Python is a programming language that lets you work quickly and integrate systems more effectively. >>> [Learn More](#)

Get Started

Whether you're new to programming or an experienced developer, it's easy to learn and use Python.

[Start with our Beginner's Guide](#)

Download

Python source code and installers are available for download for all versions!

Latest: [Python 3.14.5](#)

Docs

Documentation for Python's standard library, along with tutorials and guides, are available online.

[docs.python.org](#)

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Looking for work or have a Python related position that you're trying to hire for? Our **relaunched community-run job board** is the place to go.

[jobs.python.org](#)

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Success Stories

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Python's convenience has made it the most popular language for machine learning and artificial intelligence. Python's flexibility has allowed Anyscale to make ML/AI scalable from laptops to clusters.

Python provides convenience and flexibility for scalable ML/AI by *Dean Wampler*

Use Python for...

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- GUI Development:** [tkinter](#), [PyGObject](#), [PyQt](#), [PySide](#), [Kivy](#), [wxPython](#), [DearPyGui](#)
- AI and Machine Learning:** [PyTorch](#), [TensorFlow](#), [scikit-learn](#), [Transformers](#), [Anthropic](#), [LangChain](#)
- Scientific and Numeric:** [SciPy](#), [Pandas](#), [IPython](#)
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>>> Python Software Foundation

The mission of the Python Software Foundation is to promote, protect, and advance the Python programming language, and to support and facilitate the growth of a diverse and international community of Python programmers. >>> [Learn more](#)

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